Bulk RNA sequence analysis for TGF-β induced epithelial-mesenchymal transition(EMT) modeling in breast cancer subtypes

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1 Abstract

Epithelial-Mesenchymal transition is a biological process in case of invasive breast cancer. In this scenario, the cells lose tight cell-cell adhesion and E-Cadherin is reduced and gradually gain migratory properties and invade the surrounding tissues, and enter the bloodstream. The Transforming Growth Factor-beta (TGF-β) signaling pathway plays an important role in this transition. Using Gene Expression Omnibus, gene expression data has been evaluated, where the categorization includes Basal and HER2. After phenotypic separation by PCA, differentially expression analysis showed significantly altered genes. A correlation-based gene co-expression network revealed modular organization and highly connected hub genes. A Random Forest classifier helped to identify key predictive genes. The ODE-based mechanistic model activation of SMAD signalling, along with induction of transcription factors like SNAIL and ZEB and downregulation of E-Cadherin expression , suggesting EMT dynamics. Higher TGF-β resulted in lower peak activity and decreased steady-state response, indicating stable regulatory system.

2 Introduction

In multicellular animals, cell-cell communication must be carefully regulated to meet the needs of the organism as a whole. This is done by variety of signaling molecules that are screated or expressed on the surface of one cell and bind to the receptors of the other cells. Thus integrating and coordinating the functions of the many individual cells that make up organisms as complex as human beings.The TGF-β (Transforming Growth Factor-beta) signaling pathway helps in such cell fate decisions. One of the cellular behavior is Epithelial-mesenchymal transition(EMT). EMT plays important role in cancer cell progression. The combination of data driven statistical model and mechanistic models help to figure regulatory structure from gene expression data, identifying key drivers for phenotypic variation and translates these insigts into dynamic systems model.

3 Objective

The objective of this project to observe the dose-dependent response of TGF-β, in breast cancer cells by promoting EMT. In addition, the study aims top DEGs in distinct cellular conditions and the genes, responsible for model predictions. Furthermore, the project interprets the system follows the switching of cell fate.

4 Dataset

Gene Expression Omnibus (GEO) dataset gives gene expression data, where samples are categorized into Basal and HER2 subtypes. • Dataset ID: GSE45827 • Source: Public breast cancer gene expression dataset

5 Methods

5.1 PCA (Principal component analysis)

: Identification of dominant axes of variation and visualization of phenotypic separation was done by Principal component analysis (PCA). Different clustering of samples (Basal vs HER2) showed the underlying regulatory states.

5.2 Differential expression analysis

: A two sample t test is used,where:

- $p < 0.05$
- $|\log_2(foldchange)| > 1$

5.3 Visualization

: Using heatmaps,the differentially expressed genes presented structured expression patterns across samples.

5.4 Network inference

: Correlation techniques are applied to create co-expression network. Graph-based analysis was used to identify key nodes and modules. Edges was described by threshold correlation values, which result in a sparse network with: • Well-connected hubs (candidate regulators) • Modular structure (co-regulated gene clusters)

5.5 Feature Importance Analysis

To rank genes on their contribution to phenotypic classification, a predictive model is used.

5.6 Machine Learning Classification

A Random Forest classifier was trained to distinguish between Basal and HER2 samples: • Data split into training and test sets • Model performance evaluated using a confusion matrix • Feature importance scores used to rank genes

5.7 Mechanistic Modeling for EMT Dynamics Under TGF-β

: The Ordinary Differential Equations (ODE) model, with the crucial regulators, is created using key regulators, like SMAD, SNAIL, ZEB and E-Cadherin. The system was modeled by non-linear interaction and simulated over time by numeral integration.

5.8 TGF-β signaling

: The effect of varying TGF-β signaling capability was explored by modifying model parameters and observing system dynamics.

6 Result

6.1 PCA

PCA showed distinct clustering of samples (e.g. Basal vs HER2).

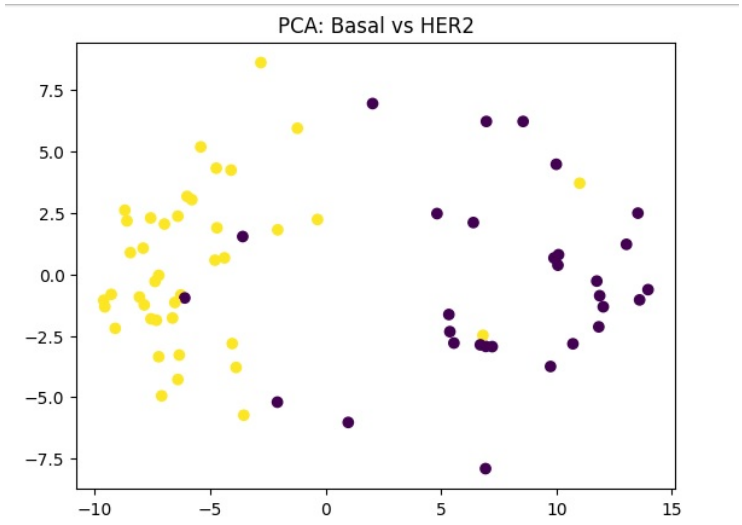

Figure 1: PCA showing separation of Basal and HER2 samples

6.2 Differential Expression

This analysis revealed a subset of genes,connected to phenotypic variation.

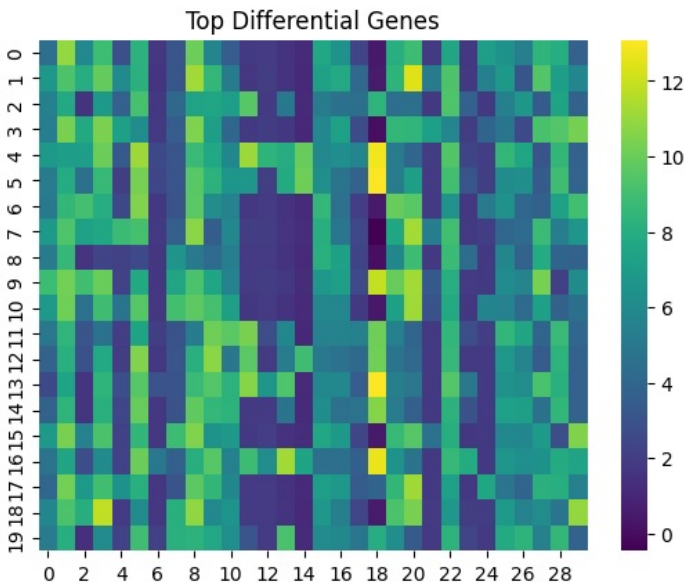

Figure 2: Top differential genes

6.3 Network Inference

The gene network revealed: • A compact core of highly connected nodes • Peripheral nodes showing low connectivity • Modular structure suggestive of functional gene clusters These features are well known characters of biological regulatory networks.

6.4 Feature Importance

It suggests key regulatory genes, which are driving the behaviour of the biological system.

6.5 Mechanistic Modeling by TGF-β

Simulation of the ODE model revealed rapid activation of SMAD, followed by moderate induction in SNAIL and ZEB signaling. It also showed repression of E-cadherin.

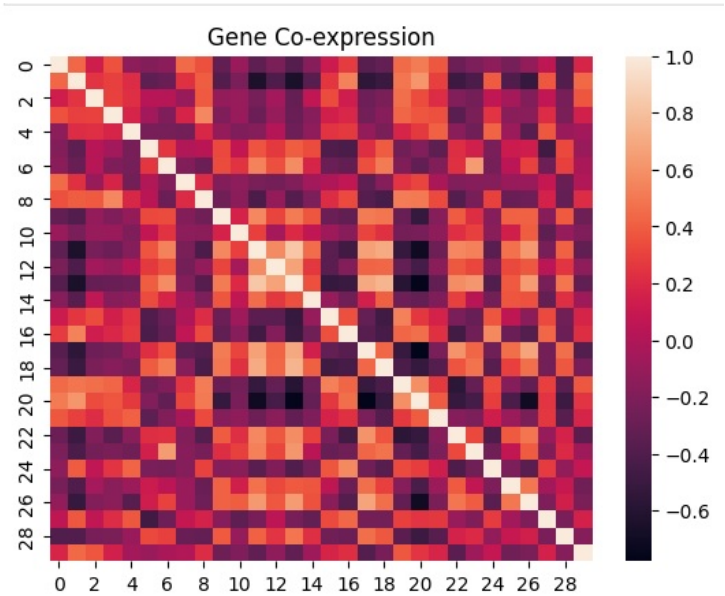

Figure 3: Gene co-expression network showing hubs and modules

6.6 Effect of TGF-β Levels

There was initial spike in activity. Then there was stabilization and it shows graded reaction. Increasing TGF-β leads to reduced peak response and lower steady state levels.

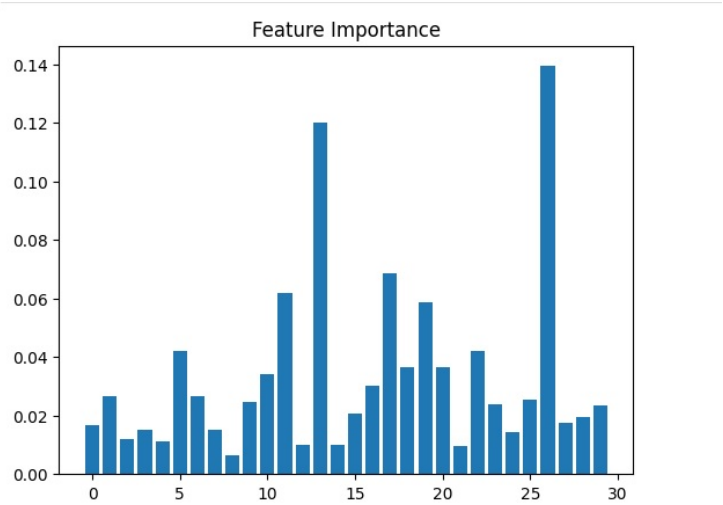

Figure 4: feature$_importance$

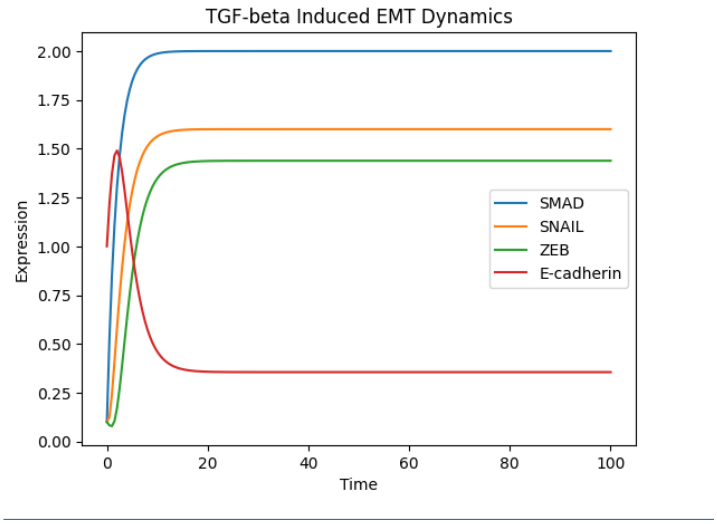

Figure 5: TGF-β induced EMT dynamics showing SMAD activation and E-cadherin repression

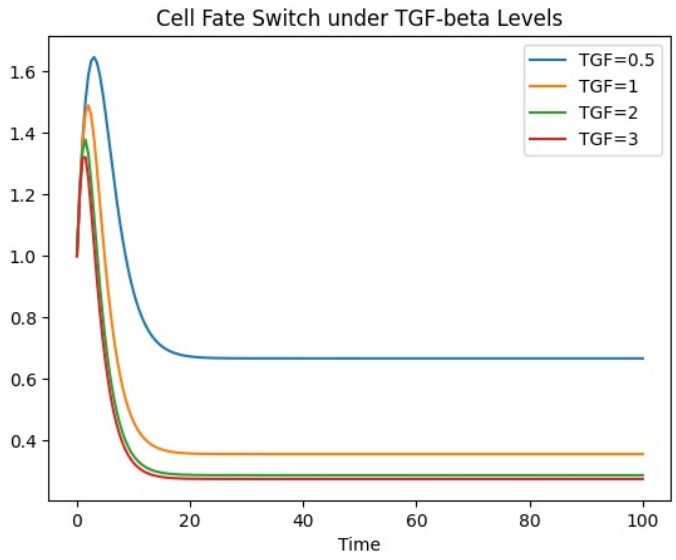

Figure 6: cell fate

7 Limitations

- Feature importance does not imply causation
- Lack of strong feedback loops
- Limited regulatory components

8 Future Work

- Introduce feedback loops for bistability
- Include additional regulators
- Experimental validation

9 Conclusion

This study integrates statistical and mechanistic approaches to model EMT dynamics. The results suggest a stable, graded response to TGF-β signaling.

10 Keywords

EMT, TGF-β, breast cancer, gene expression, PCA, Random Forest, ODE modeling

References

[1] Strasen, J. et al. *Molecular Systems Biology*, 2018.

[2] Fritzsch, C. et al. *Molecular Systems Biology*, 2018.

[3] Bohn, X. et al. *Proceedings of the National Academy of Sciences (PNAS)*, 2023.

[4] Hartmann, Y. et al. *Life Science Alliance*, 2024.